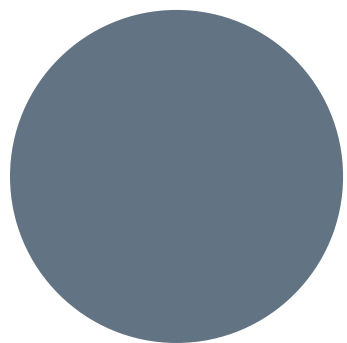
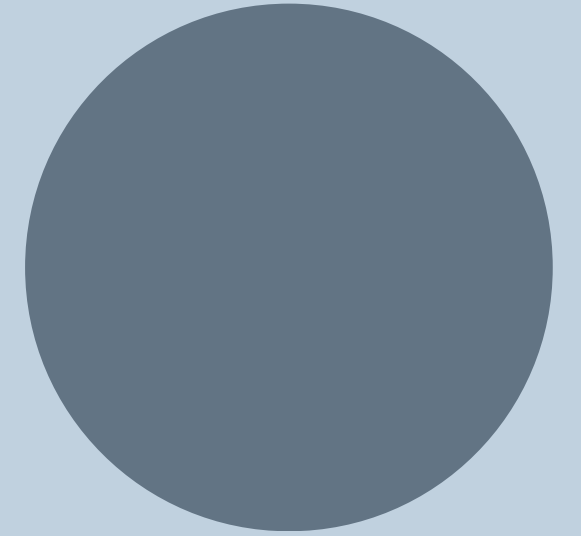

IdleInsights

Project Progress Presentation
CSE 4540: Web Programming Lab



February 18, 2025



MEET OUR TEAM



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WHAT IS IDLEINSIGHTS?

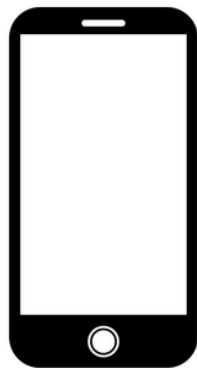
A MERN-stack self-diagnostic web app that turns procrastination from a personal failure into behavioral data science. Not a to-do app; it's a behavioral analysis system.

- **Data Capture:** User logs intended tasks and distractions (category, duration, details)
- **Pattern Recognition:** Calculate Procrastination Score and identify Root Cause Patterns (Overwhelm Avoidance, Doomscroll Loop, etc.)
- **Intervention:** Deliver personalized Countermeasure Modes to break procrastination cycles

PROJECT THEMES

● Life, But Make It Weird

- Take a boring everyday task and turn it into something hilariously dramatic or over-engineered.
- The boring task? Procrastinating itself. Everyone does it, nobody thinks twice about it



- Procrastination gets a **clinical diagnosis** with behavioral labels and personalized interventions
- "**Digital Friction Mode**" forces **breathing exercises** just to keep scrolling, over-engineered guilt

PROJECT THEMES

● Uselessly Useful

- Create a tool that solves a problem nobody asked to solve, in the most over-the-top way possible
- Nobody asked for a system that **scientifically quantifies** how lazy you are; but here it is, with async algorithms, pattern detection, and a full intervention system



The irony: The more you procrastinate, the more useful data the app collects; your laziness literally fuels the system

PROPOSAL FEEDBACK

How does IdleInsights align with the theme?

Has been explained clearly in the previous slide

How do we define asynchronous algorithm?

Our PVI calculation runs as an async function triggered by a Mongoose post-save hook. When a LogEntry is saved, the algorithm asynchronously computes the PVI impact and updates the user's profile without blocking the request

Incorporate media file handling

Users can download a PVI comparison report (their score vs. system average) as a file export

TECH STACK & AUTHENTICATION

Frontend

React 19 + Vite, React Router
Custom CSS, Axios

Backend

Node.js + Express 5,
MongoDB Atlas, Mongoose
ODM

Authentication

JWT-based (register/login),
Bcrypt password hashing,
Express auth middleware on all
protected routes

Other

Service Workers (push
notifications), Page Visibility
API

4 INPUT METHODS



Active Session Tracking

- Live timer on floating widget + quick distraction logging (category, duration)
- Session summary: total time, focus score, distraction breakdown



Periodic Check-In Notifications

- Browser push notifications during active session via service worker every 15/30 min: "Still Focused?" or "Got Distracted?"
- Auto-marks as missed after 2-min timeout



Focus Detection

- Uses Page Visibility API to detect when user leaves the app tab
- If away ≥ 5 min $\rightarrow$ prompt on return: All Distracted / All Working / Mixed / Skip



Enhanced Quick Log

- Floating purple button on all pages + Ctrl+L shortcut
- Duration presets, smart task defaults, emoji category grid
- Log a distraction in under 5 seconds from anywhere in the app

ANALYSIS ENGINE

PVI Algorithm

- Async function triggered on every log entry via Mongoose post-save hook
- Factors: distraction duration × avoided task difficulty
- Produces a 0–100 PVI score

Root Cause Pattern Detection

- Analyzes logs and assigns behavioral labels:
- Overwhelm Avoidance, Doomscroll Loop, Perfectionism Delay, Boredom Escape, Context Switching

Dashboard Visualizations

- PVI Velocity Gauge (color-coded Green/Yellow/Red)
- Activity charts, time-of-day patterns
- Benchmarking: user's PVI vs system average (downloadable report)

COUNTERMEASURE MODES

Task Deconstructor

- Counters Overwhelm Avoidance
- Break large tasks into 3–5 sub-tasks with time estimates
- Each sub-task gets its own timer; visual progress bar; completion celebrations

Digital Friction

- Counters Doomscroll Loop
- Log a monitored distraction → widget offers to track browsing time
- After configurable duration → enforced breathing pause
- Choose: "Return to Task" or "Continue Browsing"

COUNTERMEASURE MODES

Done-Over-Perfect

- Counters Perfectionism Delay
- Time-box tasks with checkpoints at 25%, 50%, 75%
- "Is this good enough?" prompts with anti-perfectionism nudges
- Forced completion at 100% of time-box

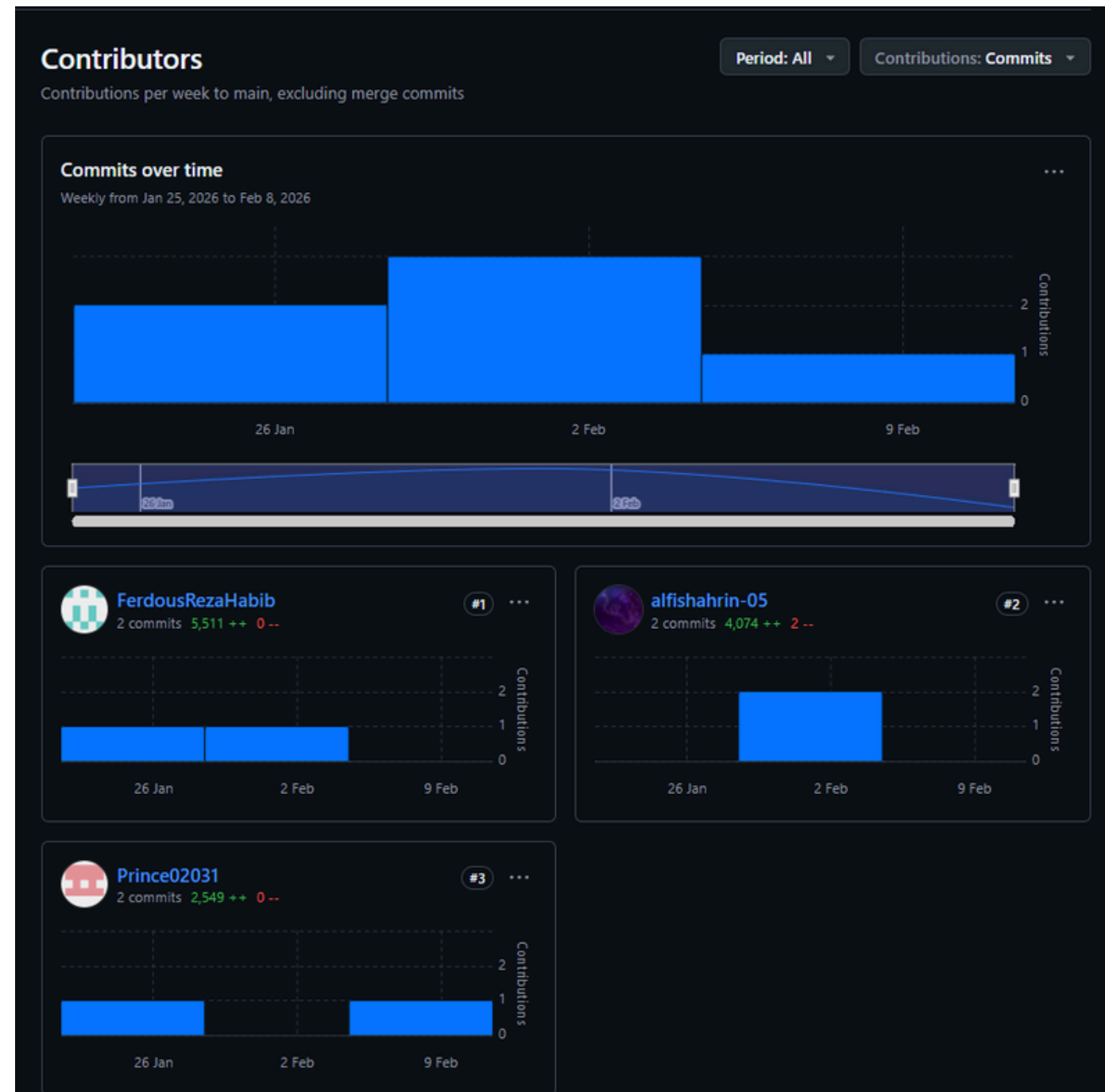
Novelty Injection

- Counters Boredom Escape
- XP system: 100 XP per 20 min focused, bonus for streaks
- Level progression (Focus Apprentice → Grindmaster)
- Random mini-challenges and achievement badges

Single Context Lock

- Counters Context Switching
- 25-min Pomodoro blocks, one task at a time
- Switch penalty warnings; "Later List" for stray thoughts
- Brain reset exercises between blocks

GITHUB CONTRIBUTION



GitHub Link of our repository
<https://github.com/FerdousRezaHabib/IdleInsights>



Live Demo



Thank You

We are open for QnA

